

eRTMAC-NWIS: Nearby Wells Intelligence System

Real-Time 3D Spatial Look-Ahead & Geomechanical Drilling Hazard Advisory System

Problem Statement ID: SIH26121

Problem Statement Title: Nearby Wells Intelligence System for Real-Time Drilling Hazard Mitigation

Ministry / Organization: Oil India Limited (Ministry of Petroleum & Natural Gas)

Theme / Category: Clean & Green Energy / Smart Automation • Software Edition

Operational Hub: eRTMAC (Real-Time Monitoring & Analytics Center, Duliajan, Assam)

Domain Calibrations: Upper Assam Shelf (Nahorkatiya, Moran, Baghjan Fields; SPE-197489)

CORE ENGINEERING HIGHLIGHTS

- ✓ 3D PostGIS Spatial Radial Query (<12ms)
- ✓ Minimum Curvature Method (MCM) 3D Trajectory
- ✓ Dip-Rotated True Stratigraphic Depth (TSD)
- ✓ Physics-Informed Teale MSE Anomaly Engine
- ✓ 1 Hz WITSML Live Telemetry Streaming
- ✓ Automated 1-Click Tour Advisory PDF Generator
- ✓ 180 / 180 Automated Rig Tests Passed (100%)

PROBLEM vs. SOLUTION ARCHITECTURE

OPERATIONAL FRICTION IN ASSAM DRILLING & eRTMAC-NWIS CAPABILITIES

[SYNTHETIC • ASSAM SPE-197489]

THE PROBLEM: SUBSURFACE UNCERTAINTY IN ASSAM

Subsurface Blind Spots: Active drilling bits enter depleted sandstones (PP 0.88 SG) and overpressured coals (PP 1.35 SG) without contextual offset awareness.

Trapped Legacy Intelligence: 50+ years of offset well records, Daily Drilling Reports (DDR), and Well Completion Reports (WCRs) are siloed in paper archives at Duliajan.

High Cost of NPT Incidents: Differential sticking in Tipam sands and reactive shale sloughing in Kopili shales cause ~35-42 hrs of Non-Productive Time (NPT), costing ₹1.8 - 4.2 Cr per event.

Geometric vs Stratigraphic Trap: Wells in anticlinal structures (Nahorkatiya) experience 3.5° SSE dip. Simple TVD matching misaligns formation horizons by up to 45m.

THE SOLUTION: eRTMAC-NWIS CAPABILITIES

3D PostGIS Offset Retrieval: Finds all offset wells within 5km radius and $\pm 200\text{m}$ TVDSS window in sub-12 milliseconds using spatial R-Tree indexing.

Dip-Rotated Stratigraphic Depth (TSD): Applies coordinate rotation along dip azimuth ($\theta=3.5^\circ$, $\phi=165^\circ$) to perfectly align geological packages.

Physics-Informed Teale MSE: Calculates real-time drilling efficiency; spikes indicate bit balling in Girujan clay or imminent sticking in depleted Tipam.

Look-Ahead Risk Index (R_H): Integrates distance-to-hazard, geological dip alignment, and historical NPT to output a 0-100% actionable risk score.

Dual-Layer User Experience: Enterprise Multi-Well Analytics Console for eRTMAC engineers + Rugged High-Contrast Doghouse Tablet Mode for rig drillers.

TECHNICAL PIPELINE & MATHEMATICAL FOUNDATION

END-TO-END DATAFLOW FROM RIG TELEMETRY TO PROACTIVE DRILLER ADVISORY

[SYNTHETIC · ASSAM SPE-197489]

01. WITSML Ingestion

1 Hz rig telemetry
ROP, WOB, Torque,
RPM,
SPP, MW, ECD, Pit Vol
Streamed via
WebSockets

02. MCM 3D Engine

Minimum Curvature
Method
Dogleg severity DL rad
Exact North, East, TVD
Continuous station
survey

03. TSD Dip Rotation

True Stratigraphic
Depth
Rotates along dip azim.
 $\Delta z = dx \sin(\theta) \cos(\phi)$
Matches faulted
horizons

04. 3D Spatial PostGIS

ST_3DDWithin query
Radius: 5km, Win:
200m
Retrieves nearby
hazards
Response time: <12ms

05. Teale MSE Engine

$MSE = WOB/Ab + 120\pi$
Torque RPM / (Ab ROP)
Physics anomaly signal
Detects balling & drag

06. Advisory & PDF

Auto Tour Advisory PDF
Generates SOP
checklists
Driller glove-touch UI
Driller sign-off record

CORE FORMULAS: Teale's MSE $[WOB/Ab + 120\pi \cdot RPM \cdot Torque / (Ab \cdot ROP)]$ · Outmans Diff-Sticking $[F_{pull} = A_c \cdot \Delta P \cdot \mu]$ · MCM 3D Trajectory

OVERCOMING HURDLES & STAKEHOLDER MATRIX

ADDRESSING TECHNICAL, GEOLOGICAL, AND ADOPTION BARRIERS IN UPSTREAM E&P

[SYNTHETIC · ASSAM SPE-197489]

STAKEHOLDER	CHALLENGES	eRTMAC-NWIS SOLUTION
Rig Drillers & Pushers Frontline operations executing drilling programs on remote rigs in Assam.	High Alert Fatigue Black-box AI models generate false alarms, leading drillers to disable or ignore warnings.	Explainable Physics (MSE) Combines Teale MSE with deterministic SOP rules so drillers know WHY an alert fires.
eRTMAC Engineers Monitoring specialists at Duliajan overseeing simultaneous multi-rig operations.	Remote Connectivity Rig sites in Upper Assam suffer intermittent WAN/cellular bandwidth to central cloud.	Edge Tablet Doghouse Mode Full local offline fallback capability with automatic reconnection to 1 Hz telemetry.
Geomechanics Team Petrophysicists managing pore pressure, mud weight windows & wellbore stability.	Stratigraphic Mismatch Geometric TVD matching fails in folded anticlines, misidentifying formation tops.	Dip-Rotated TSD Math Calculates True Stratigraphic Depth via published dip angles (SPE-197489) for precise matching.
Asset Management OIL executive leadership targeting zero-NPT campaigns and cost optimization.	Scattered Data Formats Historical well logs (LAS), DDRs (PDF/Excel), and daily lithology logs in disparate formats.	Standardized Open Schema PostGIS + TimescaleDB unified schema for WITSML, trajectory stations, and IADC hazard codes.

OPERATIONAL SCALE, FEASIBILITY & IMPACT

QUANTIFIED OPERATIONAL ROI FOR OIL INDIA LIMITED'S DRILLING CAMPAIGNS

[SYNTHETIC • ASSAM SPE-197489]

35% - 45%

REDUCTION IN NPT

Decreases differential sticking and stuck pipe incidents in depleted Tipam and Barail coals.

< 12 ms

SPATIAL QUERY TIME

PostGIS 3D R-Tree search executes across 10+ offset wells instantaneously.

₹ 14.5 Cr+

ANNUAL DRILLING SAVINGS

Prevents expensive sidetracks, fishing operations, and rig standby charges.

**100% OFF-
CLOUD**

AIR-GAPPED COMPLIANCE

Runs entirely inside OIL Duliajan intranet; zero confidential well log egress.

FIELD DEPLOYMENT ROADMAP & COMMERCIAL FEASIBILITY

Phase 1: eRTMAC Shadow Mode (Weeks 1-4): Deploy Dockerized stack (PostGIS + FastAPI) in Duliajan center; stream live WITSML from 3 active rigs in Nahorkatiya.

Phase 2: Rigsite Doghouse Pilot (Weeks 5-8): Equip drillers with rugged Android / Tauri touch tablets running offline-capable Doghouse Mode; test SOP acknowledgment.

Phase 3: Automated Tour Advisory Rollout (Weeks 9-12): Enable one-click Tour Advisory PDF generation for daily shift handovers across all Upper Assam drilling operations.

Statutory & Safety Compliance: Fully conforms to DGMS (Directorate General of Mines Safety) guidelines, OISD-GDN-178 well integrity standards, and DPDP Act 2023.

RESEARCH, REFERENCES & SCIENTIFIC VALIDATION

PEER-REVIEWED DRILLING LITERATURE, REGULATORY STANDARDS & BENCHMARKS

[SYNTHETIC · ASSAM SPE-197489]

ACADEMIC & SPE LITERATURE

SPE-197489-MS (2019): Geomechanical Modeling of Upper Assam Basin; pore pressure gradients in Barail & Tipam formations.

Teale, R. (1965): The Mechanical Specific Energy (MSE) as a Measure of Drilling Efficiency. Foundation of our anomaly engine.

Outmans, H. D. (1958): Mechanics of Differential-Pressure Sticking of Drill Collars. Calibrated pulling force equation.

SPE-166580 (2013): Real-Time Wellbore Stability & Hole Cleaning Advisory Systems in Complex Tectonic Regimes.

INDUSTRY & SAFETY STANDARDS

WITSML v1.4.1.1 & v2.0: Energetics standard for real-time rigsite data exchange between sensors and eRTMAC.

IADC Incident Ontology: Standardized drilling codes (01-23) for stuck pipe, kicks, sloughing, and BHA failures.

OISD-GDN-178 Guidelines: Oil Industry Safety Directorate guidelines for well integrity and blowout prevention.

DGMS Technical Circulars: Directorate General of Mines Safety rules governing casing pressure tests & barrier envelopes.

PROOF OF WORK BENCHMARKS

180 / 180 Tests Passed: Comprehensive 4-tier test runner (MCM, TSD, Spatial PostGIS, Teale MSE, APIs, WebSocket) achieved 100% pass in 0.119s.

Live GitHub Repository: Publicly available at github.com/sparsh101sparsh/eRTMAC-NWIS with Docker setup and reproducible seeds.

Sub-12ms Spatial Latency: PostGIS 3D spatial radial function matches offset hazards within 11.4ms average execution time.

One-Click ReportLab Engine: Instantly outputs standardized Tour Advisory PDF with full historical evidence and mitigation checklists.